Using "cin" Keyword

In C++, **cin** is an object of iostream class that is used to accept the input from the standard input stream i.e. stdin which is by default associated with keyboard. The extraction operator (>>) is used along with cin to extract the data from the object and insert it to the given variable.

Example (1) :

In this tutorial we learn to write program that input any integer number using **"cin" Keyword**.

```
#include <iostream>
using namespace std;
int main() {
    int a;
   cin >> a;
   cout << a;
   return 0;
}
```

Example (2) : This example demonstrates how to use cin in C++ programs to input a text from user input.

```
#include <iostream>
using namespace std;
int main() {
   string s;
   cin >> s;
   cout << s;
   return 0;
}
```

Example (3) :

In this program, we learn how to define variables in C++ and use them for programming in robotics, where we need to handle various types of data, from sensor readings to actuator commands.

Here we have defined four different types of variables:

```
#include <iostream>

int main() {

   // Integer variable

   int distance = 100; // Distance in centimeters

   // Floating-point variable

   float speed = 5.5; // Speed in meters per second

    // Character variable

   char direction = 'N'; // Direction as a cardinal point

   // Boolean variable

   bool is_active = true; // Status of the robot's motor

   // Output the variables

   std::cout << "Robot Status:" << std::endl;

   std::cout << "Distance: " << distance << " cm" << std::endl;

   std::cout << "Speed: " << speed << " m/s" << std::endl;

   std::cout << "Direction: " << direction << std::endl;

   std::cout << "Motor Active: " << (is_active ? "Yes" : "No") << std::endl;

   return 0;

}
```

LABORATORY TASKS:

GROUP (A),(B) and (C)

Develop the program in example (3) to allow the user to input different values at each execution for the program.